

Machine Learning Approaches in Photoplethysmography-Based Sleep Stage Classification

Tasnim Ferdous

*Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
tasnim.ferdous@bracu.ac.bd*

Reshad Ul Karim

*Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
reshad.ul.karim@g.bracu.ac.bd*

Abrar Samin

*Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
abrar.samin@g.bracu.ac.bd*

Sammam Mahdi

*Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
sammam.mahdi@g.bracu.ac.bd*

Himika Tasnim

*Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
himika.tasnim@g.bracu.ac.bd*

Aniqua Nusrat Zereen

*Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
aniqua.zereen@gmail.com*

Abstract—Multi-stage sleep classification is crucial in diagnosing sleep disorders and in evaluating sleep quality, but conventional polysomnography techniques are intrusive and time consuming. This paper examines whether continuous sleep monitoring is feasible using single-sensor-based photoplethysmography (PPG). Several machine learning classifiers including Random Forest, CatBoost, XGBoost, and SVM are used to identify the impact of sleep staging on PPG. These models can perform well in classifying between 4-sleep stages with CatBoost achieving an accuracy of 75.29% similar to the findings of the traditional PSG. The study shows that machine learning approach employed to analyze PPG signals offers a more integrated and accessible solution for the realtime sleep tracking while highlighting the importance of determining the appropriate model for the optimal performance.

Index Terms—Machine Learning, Photoplethysmography, Single sensor, Sleep Staging

I. INTRODUCTION

Polysomnography (PSG) is the traditional method for sleep staging and involves recording various physiological signals such as brain waves (EEG), muscle contractions (EMG), eye movements (EOG), heart rate, and respiratory patterns during an overnight lab stay [10]– [11]. While highly accurate, this method is labor-intensive, time-consuming, and often restricted to a single night in a sleep clinic. The shift towards telemedicine, especially during the COVID-19 pandemic, has underscored the need for less intrusive and more practical alternatives for monitoring sleep patterns [12]– [14].

Sleep is divided into four primary stages: Wake, Non-Rapid Eye Movement (NREM), which includes light and deep sleep, and Rapid Eye Movement (REM) sleep [1]– [4]. These stages play a critical role in various physiological processes, such as supporting the immune system, regulating hormones, and consolidating memory [5]– [7]. NREM sleep is characterized

by slower brain waves, while REM sleep is marked by rapid brain activity similar to wakefulness and is typically associated with vivid dreaming. Accurate differentiation between NREM and REM sleep is essential in sleep research, as disruptions in these stages, particularly due to conditions like obstructive sleep apnea (OSA), can indicate underlying neurological and metabolic disorders [8]– [9].

Photoplethysmography (PPG) has emerged as a promising alternative to PSG for sleep stage classification. PPG measures changes in blood volume within tissues, allowing for the detection of cardiovascular dynamics, breathing patterns, and heart rate variability (HRV) during sleep [15]. Wearable devices equipped with PPG sensors enable continuous and long-term sleep monitoring at home, providing a more accessible option compared to PSG [16]. However, PPG lacks direct measures of brain activity and muscle tone, which are critical for accurately distinguishing between NREM and REM stages [17]– [18].

Recent advances in machine learning (ML) have made it possible to improve the accuracy of PPG-based sleep stage classification. Several models, such as Support Vector Machines (SVM), Random Forest (RF), and more sophisticated algorithms like XGBoost and CatBoost, have been applied to the analysis of PPG data [19]– [20]. These models are particularly effective at handling the complex and nonlinear nature of the data collected from PPG sensors, making them suitable for automated sleep classification [17]– [18].

Several studies have demonstrated the effectiveness of ML models in sleep staging using PPG data. For instance, SVM models have shown a 77% accuracy in differentiating between sleep and wake stages, while Random Forest classifiers have achieved accuracy rates as high as 98% when applied to

large datasets [17]– [18]. Gradient boosting algorithms like XGBoost and CatBoost have also improved performance, with XGBoost reaching an F-1 score of 83.88% for two-stage sleep classification [19]– [20].

Despite these developments, a thorough comparison of different machine-learning strategies in the context of PPG-based sleep stage classification is still required. This study aims to fill this gap by systematically evaluating the performance of SVM, XGBoost, CatBoost, and RF classifiers on PPG data. Our work demonstrated better accuracy of 75.29% from CatBoost, 75.15% from XGBoost, 74.41% from Random Forest, and 70.20% from SVM classifier for 4-stage sleep classification from previous studies [16]. We found that CatBoost outperforms the other machine learning model for reliably and accurately classifying 4-stages of sleep. These algorithms are based on features extracted from PPG signal to deploy feasibility on wearable devices.

The proposed study has the following contributions:

- Systematic evaluation of SVM, XGBoost, CatBoost, and RF classifiers on PPG data for four sleep stage classification.
- Validation of the models on a public dataset of patients with hypopnea, showing competitive precision compared to traditional PSG-based methods.
- Statistical analysis to reduce the feature counts.

The rest of the of paper is structured in the following: The Materials and Methods section describes the details about the dataset, process flow, model development. The Results present model performance metrics, while the discussion interprets findings, limitations, and implications. The conclusion summarizes the results with a suggestion of future research directions.

II. MATERIALS AND METHODS

A. Dataset

This study included PPG data from 10 people, each recorded for a single night (7-10 hours) and PSGs collected from patients with sleep-disordered breathing at Charite Hospital, Berlin, Germany [21]. The PSG recordings are divided into 30-second epochs and labeled to six stages according to Rechtschaffen and Kales (R & K).

R&K uses a more granular approach with six stages, splitting light sleep into S1 and S2, and deep sleep into S3 and S4. Our data set follows classic R & K scoring and the purpose of this study is to identify four sleep stages. Therefore S1 and S2 are classified as Light Sleep, and S3 and S4 as Deep Sleep. Therefore, the light and deep Sleep stages can be compared with the Standard AASM scoring system, where deep sleep is classified in one stage (N3) and light sleep into N1 and N2. Table I summarizes the subject-wise segmentation of annotated PPG data based on their Apnea- Hypopnea Index (AHI) score.

B. Data Pre-processing

The PPG data, originally captured at 128 Hz, is down-sampled to 64 Hz to improve computation speed and

TABLE I. SUBJECT-WISE 4-SLEEP STAGE [WAKE-LIGHT-DEEP-REM] ANNOTATION DISTRIBUTION

Subject No.	Total Segments	Wake	Light	Deep	REM
Subject 1	1246	629	439	142	36
Subject 2	1212	795	214	87	116
Subject 3	924	245	331	243	105
Subject 4	909	388	288	134	99
Subject 5	909	202	434	221	52
Subject 6	872	192	482	108	90
Subject 7	864	102	448	177	137
Subject 8	844	108	608	107	21
Subject 9	833	500	168	105	60
Subject 10	815	247	356	121	91
Total	9428	3408	3768	1445	807

reduce model complexity, focusing on capturing general sleep patterns rather than fine-grained details [22]. The dataset contains 9,428 sleep epochs, each 30-second intervals. The segment lengths ranging from 815 to 1,246, and 84 characteristics are standardized using Z-score after detrending [21]. By statistical analysis we reduced the features to 72, focusing on the features with similar value which spreaded throughout the column are removed. The standard score Z is calculated by subtracting the mean μ of the sample from the observed value x , and then dividing the result by the standard deviation σ of the sample:

$$Z = \frac{x - \mu}{\sigma} \quad (1)$$

C. Process flow: Machine Learning approach

The data is then split into two different approaches for training and testing. Approach - 1: 70% of the data allocated for training and 30% for testing on 9428 samples where 6660 samples are utilized for training and 2828 for testing. Approach - 2: Subject-wise 70% for training and 30% for testing as shown in Fig. 1.

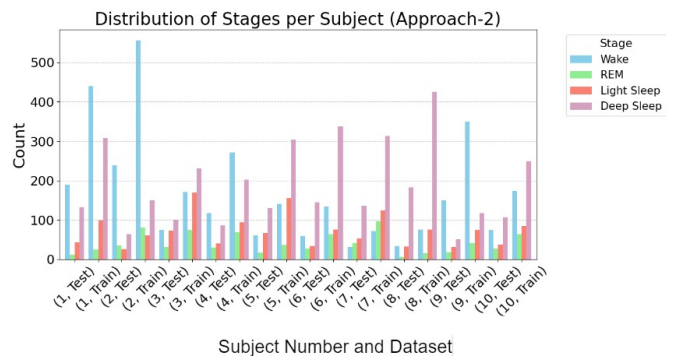


Fig. 1. Dataset Distribution of Approach - 2.

The segmented data undergoes various machine learning classifiers, such as SVM, RF, CatBoost and XGBoost to identify the optimal model. The Best Model is selected based on its performance, classifying the data into four sleep stages: Wake, Light, Deep, and REM. Fig. 2 depicts the overall process flow diagram for the data training and testing.

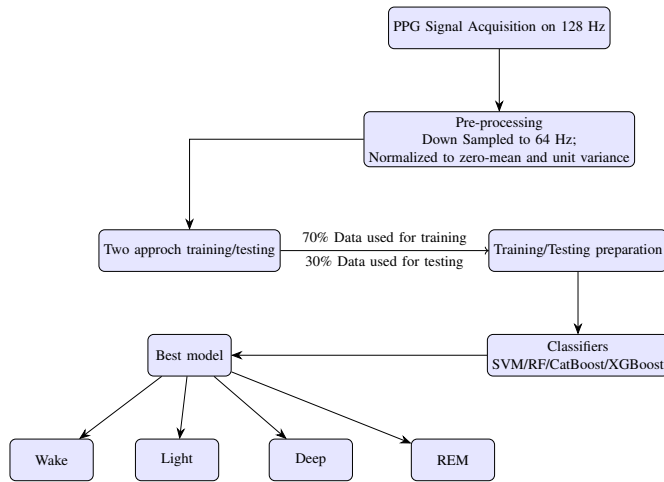


Fig. 2. Complete sleep stage classification Process flow for 4-stage sleep classification.

D. Model Initialization and Training with Default Parameters

Four machine learning models— RF, SVM, XGBoost, and CatBoost are selected for the classification task. Each model is initially trained using its default hyperparameters to establish baseline performance metrics.

- 1) *Random Forest*: The RF is initialized and trained using 300 trees ‘n_estimators=300’ with other parameters set to their defaults.
- 2) *Support Vector Machine*: The SVM model is wrapped in a pipeline that included data scaling using ‘StandardScaler’ to ensure that all features contributed equally to the model’s decision-making process. The SVM is trained using its default settings.
- 3) *XGBoost Classifier*: The XGBoost model is initialized and trained without any specific hyperparameter adjustments, relying on its internal mechanisms to handle imbalanced classes and feature interactions.
- 4) *CatBoost Classifier*: Similarly, CatBoost is trained using its default parameters, leveraging its gradient boosting approach optimized for categorical data.

The models are evaluated on the test set, and key performance metrics such as accuracy scores, classification reports, and ROC curves are generated for each model.

E. Hyperparameter Tuning

To further enhance the performance of the models, extensive hyperparameter tuning is conducted using both grid search and randomized search techniques. The goal is to identify the optimal configurations that will yield the best classification model.

- 1) *Random Forest*: The model is configured with the following hyperparameters: *Number of Trees* is set to 600, increasing the ensemble size to capture more patterns. The *Max Depth* is set to 20, allowing the model to grow deeper trees to capture complex relationships. The *Min Samples Split* is set to 5, controlling the

minimum number of samples required to split a node. Additionally, *Bootstrap* is disabled (‘bootstrap = False’), ensuring all data is used to grow each tree.

- 2) *Support Vector Machine*: The model is set with the following hyperparameters: The *Kernel* is set to ‘rbf’ to capture non-linear relationships. The *C* parameter is set to 100.0, increasing the model’s tolerance for misclassification on the training data. *Class Weight* is balanced to address class imbalances. Additionally, *Probability* is enabled to allow for probability estimates, facilitating ROC curve generation.
- 3) *CatBoost Classifier*: The model is configured with the following hyperparameters: *Bagging Temperature* is set at 0.4175 which controls randomness of bagging and influences the diversity of individual trees. *Boosting Type* is configured as ‘Ordered’ allowing the use of ordered boosting, which is beneficial for categorical features. The *Bootstrap Type* is configured ‘MVS’ that defines the method used for data sampling for building trees. The *Depth* is set to 9 which enables the model to capture complex patterns by creating deeper trees. Moreover, *Iterations* are set to 858 which finds out the number of boosting iterations. The *Learning Rate* is configured to 0.2898 to balance the speed and accuracy of learning. Additionally, *Class Weights* are custom-assigned to handle class imbalance effectively.
- 4) *XGBoost Classifier*: The model is fine tuned with the following hyperparameters: The *Objective* is defined as ‘multi:softmax’ for multi-class classification. The *Max Depth* value is 9 to control the complexity of the model. The *Learning Rate* is configured at 0.05 to ensure gradual convergence. Then the *Subsample* is tuned to 0.9 which controls the fraction of samples used to grow trees. Regularization Parameters are set as follows: ‘reg_lambda’ is set to 0.04 and ‘reg_alpha’ is set to 0.7 which helps in managing overfitting.

The hyperparameters are fine-tuned empirically, and the optimal configurations are determined through a combination of random search. This approach ensured that each model is tailored to the specific characteristics of the dataset.

F. Model Evaluation and Analysis

Following hyperparameter tuning, the models are re-evaluated on the test set. The performance metrics—classification reports, accuracy scores, and ROC curves—are generated again to compare the results with the baseline. Additionally, confusion matrices are produced for each model to provide a detailed understanding of the false positives, false negatives, true positives, and true negatives. This analysis is very important in finding out the strengths and weaknesses of each of the models in terms of precision, recall and overall classification accuracy. Equation [2-7] are utilized to evaluate it.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2)$$

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (3)$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad (4)$$

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (5)$$

$$p_o = \frac{\text{Correctly Identified Test Cases}}{\text{Total Test Cases}} \quad (6)$$

$$p_e = \sum_{i=1}^k \frac{(TP_i + FP_i) \cdot (TP_i + FN_i)}{n^2} \quad (7)$$

III. RESULT

A. Performance Evaluation

In this study, the performance of four different machine learning models such as RF, SVM, XGBoost, and CatBoost—is assessed thoroughly for multi-class classification. Each model is examined using multiple metrics which includes accuracy, precision, recall and F1-score and further insights are achieved by analysing the ROC curves and confusion matrices.

- 1) *RF*: Recorded 74.41% accuracy with precision (0.70-0.80), recall (0.40-0.84), and F1-score (0.53-0.81). Strong in the REM class, but lower recall for the Deep sleep class.
- 2) *SVM*: Lower accuracy at 70.20% with precision (0.58-0.78), recall (0.66-0.76), and F1-score (0.62-0.77). Struggled with the Wake class.
- 3) *XGBoost*: Achieved 75.15% accuracy with precision (0.71-0.79), recall (0.45-0.85), and F1-score (0.57-0.82). Excelled in classifying REM.
- 4) *CatBoost*: Matched XGBoost with 75.29% accuracy, precision (0.73-0.83), recall (0.50-0.86), and F1-score (0.62-0.81). Particularly effective in the REM. Table II shows The summary of accuracy, precision, recall and F1-score of each model without hyperparameter tuning and Table III shows the same summary with hyperparameter tuning.

B. Confusion Matrix

The confusion matrices for the RF, SVM, XGBoost and CatBoost models show great overall classification performance, with each model performing well in certain classes but faltering in others. XGBoost in Fig. 3 shows

TABLE II. PERFORMANCE METRICS OF DIFFERENT MODELS WITHOUT HYPERPARAMETER TUNING

Model	Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)
RF	72.50	75	63	66
SVM	66.21	73	52	55
XGBoost	74.80	76	67	70
CatBoost	74.34	75	67	70

TABLE III. PERFORMANCE METRICS OF DIFFERENT MODELS WITH HYPERPARAMETER TUNING

Model	Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)
RF	74.41	77	66	69
SVM	70.20	67	70	68
XGBoost	75.15	76	67	70
CatBoost	75.29	77	69	72

particular difficulty in discriminating between classes LS and REM, whereas SVM in Fig. 4 performs well but frequently confuses between class LS with class Wake. The RF model in Fig. 5 is very accurate in classes LS and Wake, but performs moderately in classes DS and REM, with some misclassifications between comparable classes. CatBoost provides a balanced performance across all classes, with substantial accuracy in classes LS and REM which is evident in Fig. 6.

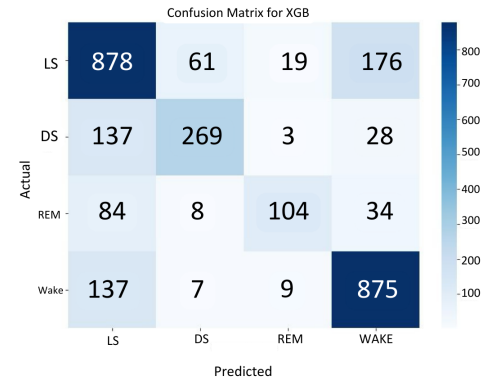


Fig. 3. XGBoost Confusion Matrix.

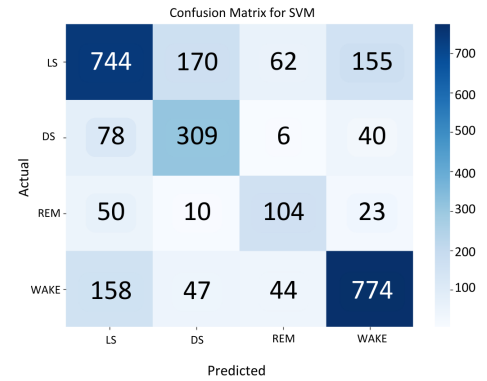


Fig. 4. SVM Confusion Matrix.

C. ROC Curve Comparison

The comparison of ROC curves with and without hyperparameter tuning demonstrates the importance of fine-tuning on classification performance. Fig. 7 shows that hyperparameter adjustment increases the AUC for XGBoost, RF and CatBoost to 0.93, and SVM to 0.91, indicating improved discriminatory power. In comparison, Fig. 8 indicates that untuned models had slightly lower AUCs, with XGBoost, Random Forest, and CatBoost at 0.92 and SVM at 0.88. The tuned ROC curves are smoother, showing more

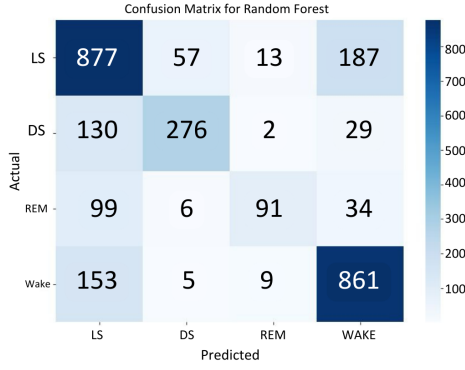


Fig. 5. Random Forest Confusion Matrix.

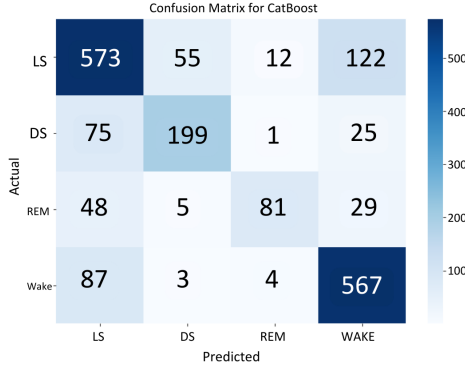


Fig. 6. CatBoost Confusion Matrix.

consistent true positive rates across thresholds, especially in the SVM model, which exhibits a significant improvement in performance after tuning. Overall, hyperparameter adjustment improves model performance, as evidenced by improved AUC values and optimized ROC curves, which improve classification accuracy across different thresholds.

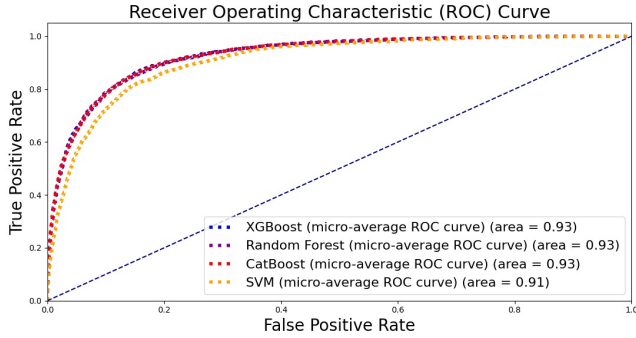


Fig. 7. ROC Curve with Hyperparameter Tuning.

IV. DISCUSSION

Our study depicted in Table IV, shows that all of our classifiers achieved better accuracy and F-1 score for a four stage sleep classification compared to previous works [16], [23]. The empirical analysis of tuning hyperparameters in the selected models and statistical analysis of choosing features outperformed all the existing research.

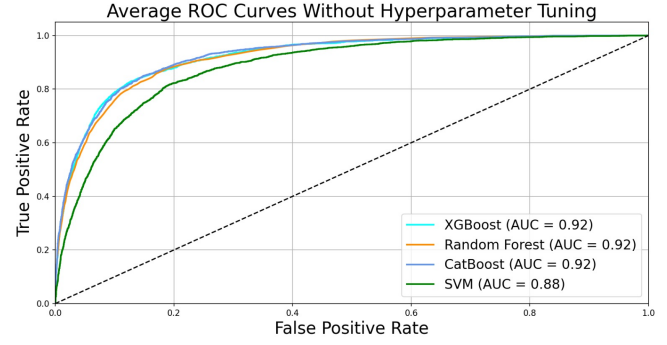


Fig. 8. ROC Curve without hyperparameter tuning.

The small data set of 10 patients with hypopnea in this study is one of its limitations, which may have an impact on how effectively the models generalize to new data. Data augmentation could be used to improve this restriction. It was not employed in this work, as for a small dataset it might produce artificial patterns that do not accurately reflect the true variability of PPG data in real-world scenarios. Another limitation addresses the use of 72 features to obtain the accuracy of PSG. However, large number of features increases the possibility of overfitting and adds computational complexity, which traditional machine-learning models may not handle efficiently. Furthermore, compared to deep learning techniques, the exclusive use of machine learning models can make it harder to identify complex patterns in PPG data.

In future research, Explainable AI (XAI) approaches to enhance performance and interpretability, as well as lower dimensionality and improve feature selection, could be used to overcome these restrictions. XAI could assist in identifying the most relevant features, allowing for more efficient data management and lower computing expenses. Furthermore, combining deep learning models with transfer learning from bigger datasets may help alleviate the problems associated with small dataset sizes by exploiting pre-trained knowledge to increase model accuracy and robustness.

V. CONCLUSION

In this study, we evaluated multiple machine learning models for four-stage sleep classification using PPG data, with CatBoost achieving the greatest accuracy of 75.29% and F1-Score of 0.72, exhibiting competitive performance against state-of-the-art approaches. While CatBoost and other traditional machine learning models are easier to develop and have lower computing costs, their small dataset size and high feature dimensionality can restrict generalizability and resilience.

For single-sensor systems like PPG, machine learning can achieve promising results despite the challenges. Enhancing feature selection and exploring advanced techniques like transfer learning could enhance model performance and better capture the complex patterns present in PPG data. By refining these models, more accurate and efficient sleep monitoring systems could be developed for real-world applications.

TABLE IV. SUMMARY OF STATE OF THE ART OBTAINED RESULTS FOR THE 4 STAGE [WAKE-LIGHT-DEEP-REM] SLEEP CLASSIFICATION

	Accuracy(%)	Precision (%)	Recall (%)	F1-Score(%)	κ	Dataset	Method
Wu et al.[2020] [23]	62	-	-	-	0.41	Private(N=31)	PPG with SpO ₂ , 55 features, SVM-ANN
Motin et al.[2023] [16]	70.19 $\pm$ 0.32	-	-	62	-	Private(N=10)	PPG, 72 features, SVM
Motin et al.[2023] [16]	72.39 $\pm$ 0.48	-	-	62	-	Private(N=10)	PPG, 72 features, RF
Motin et al.[2023] [16]	72.23 $\pm$ 0.38	-	-	65	-	Private(N=10)	PPG, 72 features, KNN
Proposed Model	75.29	77	69	72	0.58	Private(N=10)	PPG, 72 features, CatBoost
Proposed Model	75.15	76	67	70	0.58	Private(N=10)	PPG, 72 features, XGBoost
Proposed Model	74.41	77	66	69	0.57	Private(N=10)	PPG, 72 features, RF
Proposed Model	70.20	67	70	68	0.50	Private(N=10)	PPG, 72 features, SVM

REFERENCES

- [1] M. P. Walker and E. van der Helm, "Overnight therapy? The role of sleep in emotional brain processing," *Psychological Bulletin*, vol. 135, no. 5, pp. 731–748, 2009.
- [2] B. Rasch and J. Born, "About sleep's role in memory," *Physiological Reviews*, vol. 93, no. 2, pp. 681–766, 2013.
- [3] J. A. Hobson, R. W. McCarley, and P. W. Wyzinski, "Sleep cycle oscillation: Reciprocal discharge by two brainstem neuronal groups," *Science*, vol. 189, no. 4196, pp. 55–58, 1975.
- [4] M. A. Carskadon and W. C. Dement, "Chapter 2 - Normal human sleep: An overview," in *Principles and Practice of Sleep Medicine*, M. H. Kryger, T. Roth, and W. C. Dement, Eds. Philadelphia, PA: Elsevier Saunders, 2011, pp. 16–26.
- [5] R. Stickgold and M. P. Walker, "Sleep-dependent memory triage: Evolving generalization through selective processing," *Nature Neuroscience*, vol. 16, no. 1, pp. 139–145, 2013.
- [6] A. N. Vgontzas, E. O. Bixler, and G. P. Chrousos, "Metabolic disturbances in obesity versus sleep apnea: The importance of visceral obesity and insulin resistance," *Journal of Internal Medicine*, vol. 254, no. 1, pp. 32–44, 2003.
- [7] M. R. Irwin, "Why sleep is important for health: A psychoneuroimmunology perspective," *Annual Review of Psychology*, vol. 66, pp. 143–172, 2015.
- [8] K. A. Franklin and E. Lindberg, "Obstructive sleep apnea is a common disorder in the population—a review on the epidemiology of sleep apnea," *Journal of Thoracic Disease*, vol. 7, no. 9, pp. 1311–1322, 2015.
- [9] A. Iranzo, J. Santamaria, and E. Tolosa, "Idiopathic rapid eye movement sleep behavior disorder: Diagnosis, management, and the need for neuroprotective interventions," *The Lancet Neurology*, vol. 15, no. 4, pp. 405–419, 2016.
- [10] C. Iber, S. Ancoli-Israel, A. L. Chesson, and S. F. Quan, "The new sleep scoring manual – The evidence behind the rules," *Journal of Clinical Sleep Medicine*, vol. 3, no. 2, pp. 107, 2007.
- [11] A. Kales and A. A. Rechtschaffen, *Manual of Standardized Terminology, Techniques and Scoring System for Sleep Stages of Human Subjects*. Washington, DC: U.S. Government Printing Office, 1968.
- [12] R. B. Berry, R. Brooks, C. E. Gamaldo, S. M. Harding, R. M. Lloyd, and S. F. Quan, "AASM scoring manual updates for 2017 (version 2.4)," *Journal of Clinical Sleep Medicine*, vol. 13, no. 5, pp. 665–666, 2017.
- [13] R. N. Sekkal, S. Belala, K. Nader, and A. Belouchrani, "Automatic sleep stage classification: From classical machine learning methods to deep learning," *Biomedical Signal Processing and Control*, vol. 77, Art. no. 103751, 2022.
- [14] K. G. Johnson, A. Sullivan, and J. N. Chervin, "The impact of the COVID-19 pandemic on sleep medicine practices," *Journal of Clinical Sleep Medicine*, vol. 17, no. 1, pp. 79–87, 2021.
- [15] Y. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA: MIT Press, 2016.
- [16] M. A. Motin, L. A. Lewis, P. J. Greenwood, and J. M. Allen, "Multi-stage sleep classification using photoplethysmographic sensor," *Royal Society Open Science*, vol. 10, no. 4, Art. no. 221517, 2023.
- [17] E. Smily Jeya Jothi, J. Anitha, and D. Hemanth, "An automatic screening approach for obstructive sleep apnea from photoplethysmograph using machine learning techniques," *IEEE Transactions on Instrumentation and Measurement*, vol. 70, pp. 1–8, 2021.
- [18] M. K. Uçar, A. Yurtman, and C. K. Yavuz, "Automatic sleep staging in obstructive sleep apnea patients using photoplethysmography, heart rate variability signal and machine learning techniques," *Neural Computing and Applications*, vol. 29, no. 1, pp. 1–16, 2018.
- [19] R. Huttunen, K. Korkalainen, H. Kainulainen, H. Korkalainen, and T. Leppänen, "Assessment of obstructive sleep apnea-related sleep fragmentation utilizing deep learning-based sleep staging from photoplethysmography," *Sleep*, vol. 44, no. 6, Art. no. zsab142, 2021.
- [20] D. A. Almeida, P. J. Pereira, and H. J. Martins, "A machine-learning sleep-wake classification model using a reduced number of features derived from photoplethysmography and activity signals," *ArXiv*, abs/2308.05759, 2023.
- [21] A. Habib, L. Al-Fahoum, A. Al-Smadi, and D. L. Bester, "Performance of a convolutional neural network derived from PPG signal in classifying sleep stages," *IEEE Transactions on Biomedical Engineering*, vol. 70, no. 6, pp. 1717–1728, June 2023.
- [22] H. Korkalainen, T. Leppänen, K. Kainulainen, J. Himanen, and M. Töyräs, "Deep learning enables sleep staging from photoplethysmogram for patients with suspected sleep apnea," *Sleep*, vol. 43, no. 11, pp. 1–10, 2020.
- [23] X. Wu, X. Chen, Y. Song, and H. Luo, "Automatic sleep-stage scoring based on photoplethysmographic signals," *Physiological Measurement*, vol. 41, no. 6, Art. no. 065008, June 2020.